

Product Form, Time Reversal, and External Departure Flows in an Open Jackson Network

Route-A source-local certificate HCS-C351

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Abstract

For a finite open network of single-server exponential queues, we prove the exact traffic solution, product-geometric invariant law, and stability criterion in one fixed row-vector convention. We compute every parameter of the stationary reversed Jackson network, including the routing-index reversal and nodes without direct exits.

1 Convention and theorem

Let $V = \{1, \dots, d\}$. Independent external Poisson streams have rates $\alpha_i > 0$, and node i has one exponential server of rate $\mu_i > 0$. After service at i , a customer is routed to j with probability p_{ij} , or exits with

$$p_{i0} = 1 - \sum_j p_{ij} \geq 0.$$

Assume the substochastic routing matrix $P = (p_{ij})$ satisfies $\text{spr}(P) < 1$. Self-routing is allowed, but it is a phantom service mark for the queue-length vector $X \in \mathbb{Z}_+^d$. The visible transitions are

$$\begin{aligned} x &\longrightarrow x + e_i && \text{at rate } \alpha_i, \\ x &\longrightarrow x - e_i && \text{at rate } \mu_i p_{i0} \mathbf{1}_{\{x_i > 0\}}, \\ x &\longrightarrow x - e_i + e_j && \text{at rate } \mu_i p_{ij} \mathbf{1}_{\{x_i > 0\}}, \quad i \neq j. \end{aligned} \tag{1}$$

Vectors are rows. Thus

$$\lambda = \alpha + \lambda P = \alpha(I - P)^{-1}, \quad \rho_i = \lambda_i / \mu_i. \tag{2}$$

Theorem 1. *The chain is nonexplosive and irreducible. It is positive recurrent if and only if $\rho_i < 1$ for every i . In that case its unique invariant law is*

$$\pi(x) = \prod_{i=1}^d (1 - \rho_i) \rho_i^{x_i}. \tag{3}$$

Its stationary reversal is a Jackson network, in the natural extension allowing zero exogenous rates, with service rates μ_i , traffic vector λ , and

$$\hat{\alpha}_i = \lambda_i p_{i0}, \quad \hat{p}_{ji} = \frac{\lambda_i p_{ij}}{\lambda_j}, \quad \hat{p}_{i0} = \frac{\alpha_i}{\lambda_i}. \tag{4}$$

The inverse in (2) is $\sum_{m \geq 0} P^m$, so λ exists uniquely and $\lambda_i \geq \alpha_i > 0$. The total visible jump rate is bounded by $\sum_i (\alpha_i + \mu_i)$. Also $\text{spr}(P) < 1$ makes the routing chain eventually exit from every node. A prescribed finite sequence of services, with no intervening arrivals, reaches zero with positive probability; positive external rates then reach any state. This proves nonexplosion and irreducibility.

2 Global balance and the product law

Assume $\rho_i < 1$, let $E(x) = \{i : x_i > 0\}$, and put $w(x) = \prod_i \rho_i^{x_i}$. The visible rate out of x is

$$q_{\text{out}}(x) = \sum_i \alpha_i + \sum_{j \in E(x)} \mu_j (1 - p_{jj}). \tag{5}$$

After division by $w(x)$, incoming external departures from $x + e_i$ contribute

$$\sum_i \rho_i \mu_i p_{i0} = \sum_i \lambda_i p_{i0} = \sum_i \alpha_i, \quad (6)$$

where the last equality follows by summing (2). For an occupied destination j , its incoming external arrival and all routed arrivals from $i \neq j$ contribute

$$\frac{\alpha_j}{\rho_j} + \frac{1}{\rho_j} \sum_{i \neq j} \lambda_i p_{ij} = \frac{\lambda_j(1 - p_{jj})}{\rho_j} = \mu_j(1 - p_{jj}). \quad (7)$$

Equations (6)–(7) reproduce (5), hence $wQ = 0$ at every state. Normalizing each geometric factor proves (3); irreducibility gives uniqueness and positive recurrence.

This is the *round zero product form closure*: the continuum theorem is proved by global balance, not by truncating the state space.

3 The reversed network

For a stationary forward jump $x \rightarrow y$, $\hat{q}(y, x) = \pi(x)q(x, y)/\pi(y)$. The three visible pairs give

$$\begin{aligned} x \rightarrow x + e_i & : \hat{q}(x + e_i, x) = \alpha_i / \rho_i = \mu_i \hat{p}_{i0}, \\ x \rightarrow x - e_i & : \hat{q}(x - e_i, x) = \rho_i \mu_i p_{i0} = \hat{\alpha}_i, \\ x \rightarrow x - e_i + e_j & : \hat{q}(x - e_i + e_j, x) = \mu_i p_{ij} \rho_i / \rho_j = \mu_j \hat{p}_{ji}. \end{aligned} \quad (8)$$

Thus the index order in (4) is forced: forward $i \rightarrow j$ becomes reverse $j \rightarrow i$. For reverse source j ,

$$\hat{p}_{j0} + \sum_i \hat{p}_{ji} = \frac{\alpha_j + \sum_i \lambda_i p_{ij}}{\lambda_j} = 1.$$

The reversed traffic check is equally explicit:

$$\hat{\alpha}_i + \sum_j \lambda_j \hat{p}_{ji} = \lambda_i p_{i0} + \lambda_i \sum_j p_{ij} = \lambda_i.$$

Since $\hat{p}_{j0} = \alpha_j / \lambda_j > 0$, the reversed routing matrix has every row sum below one. The forward model retains $\alpha_i > 0$, whereas formula (4) may have $\hat{\alpha}_i = 0$ when $p_{i0} = 0$; the natural zero-input extension is the convention used for the reversed network, and openness and the traffic equations remain valid.

This is the *round one reversal closure*.

Source boundary

This is a source-local reconstruction, not a priority claim. Jackson owns the open-network product-form lineage; Burke owns the classical output theorem; Kelly supplies the time-reversal and quasi-reversibility lineage. The exact software receipts are local implementation checks, not attributions to those sources.

References

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